

Sean Luka Girgis

Senior Data Engineer | Python, SQL, ETL/ELT, Relational Databases & Data Quality

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, Oracle, relational databases, PySpark/Spark, ETL/ELT pipelines, data warehouses, data marts, data quality, operational support, monitoring, documentation, Agile delivery, and governance-aware enterprise reporting. Strong background designing and supporting data workflows that ingest from databases, files, APIs, and monitoring platforms; transform raw operational data into trusted analytics datasets; optimize SQL/reporting layers; and support incident management for production data flows. Best aligned to enterprise data engineering roles requiring Python, SQL, relational databases, ETL/ELT, data modeling, operational support, monitoring, documentation, mentoring, and secure/scalable delivery; Azure SQL, Azure Data Factory, Synapse, Cosmos DB, Azure DevOps, T-SQL, SSIS/Informatica/Kafka/MuleSoft are positioned carefully as transferable or adjacent areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark and SQL ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS and warehouse-backed data platform components using S3, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and downstream analytics.
- Developed data extraction, transformation, and loading workflows from monitoring platforms, relational databases, files, and operational feeds into reporting datasets, data marts, and analytics-ready structures.
- Implemented data quality and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve accuracy, completeness, consistency, and trust in outputs.
- Optimized complex SQL queries, Oracle schemas, Redshift workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Provided operational support and incident investigation for data pipelines and reporting flows, including source-feed issues, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Documented data flows, source mappings, metric definitions, transformation rules, validation checkpoints, runbooks, support procedures, and technical designs to improve visibility and reuse.
- Partnered with infrastructure, analytics, application, engineering, governance, and business teams to refine requirements, define acceptance criteria, explain issues, and deliver reliable enterprise data assets.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Investigated real-user and synthetic monitoring source data to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by application, middleware, infrastructure, and operations teams.
- Provided troubleshooting guidance, onboarding support, training material, and knowledge sharing to improve adoption of monitoring and reporting standards.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Automated schema conversion, data validation, phased cutover support, troubleshooting, and production-readiness activities for a mission-critical migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java, Scala concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and issue documentation.

Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting workflows.

Technologies: Python, Pandas, PySpark, SQL, Prophet, Data Validation